

1 Problem formulation

Low-order representation A low-order representation of spatiotemporal data $q(\mathbf{x}, t)$ can be written as

$$q(\mathbf{x}, t) \simeq \sum_{k=1}^K \phi_k(\mathbf{x}) c_k(t), \quad (1.1)$$

where ϕ_k is the (spatial) mode and c_k is the temporal coefficient. The data $q(\mathbf{x}, t)$ can be either real- or complex-valued.

RVMD modes RVMD seeks a group of triplets with a finite number K

$$\{\phi_k(\mathbf{x}), c_k(t), \omega_k\}_{k=1}^K \quad (1.2)$$

that constitutes an approximation of the observed data. We define the inner products and the induced norms as below

$$\|\phi(\mathbf{x})\|_{\mathbf{x}, W} \equiv \langle \phi, \phi \rangle_{\mathbf{x}, W}^{1/2}, \quad \langle \phi_1, \phi_2 \rangle_{\mathbf{x}, W} \equiv \int_{\Omega} \phi_1(\mathbf{x}) \overline{\phi_2(\mathbf{x})} W(\mathbf{x}) d\mathbf{x} \quad (1.3)$$

$$\|c(t)\|_t \equiv \langle c, c \rangle_t^{1/2}, \quad \langle c_1, c_2 \rangle_t \equiv \int_{\Omega} c_1(\mathbf{x}) \overline{c_2(\mathbf{x})} d\mathbf{x} \quad (1.4)$$

where $W(\mathbf{x})$ is the weight function defined on spatial domain. Domains of $\phi(\mathbf{x})$, $c_k(t)$, and ω_k are

$$\Phi = \{\phi(\mathbf{x}) \in L^{2, \mathbb{C}} \mid \|\phi(\mathbf{x})\|_{\mathbf{x}, W} = 1\}; \quad C = L_{loc}^{2, \mathbb{C}}(-\infty, \infty); \quad \mathbb{R}^+, \quad (1.5)$$

respectively.

The RVMD optimization problem The optimization can be formulated in the frequency domain

$$\min_{\{\phi_k(\mathbf{x}), \hat{c}_k(\omega), \omega_k\}_{k=1}^K} \left\{ \sum_{k=1}^K \alpha \int_{-\infty}^{\infty} (|\omega| - \omega_k)^2 |\hat{c}_k(\omega)|^2 d\omega + \int_{\Omega} \int_{-\infty}^{\infty} \left| \hat{q}(\mathbf{x}, \omega) - \sum_{k=1}^K \phi_k(\mathbf{x}) \hat{c}_k(\omega) \right|^2 W(\mathbf{x}) d\omega d\mathbf{x} \right\}. \quad (1.6)$$

Here $\hat{c}_k(\omega) \equiv \mathcal{F}\{c_k(t)\} \in \hat{C} \equiv L_{loc}^{2, \mathbb{C}}(-\infty, \infty)$.

2 Computation

Using block coordinate descent algorithm to compute RVMD modes, the original joint problem is then decomposed into a sequence of iterative sub-optimizations

$$\phi_k^{n+1} = \operatorname{argmin}_{\phi_k \in \Phi} \{F_{\phi_k}^{n+1}(\phi_k)\}, \quad (2.1)$$

$$\hat{c}_k^{n+1} = \operatorname{argmin}_{\hat{c}_k \in \hat{C}} \{F_{\hat{c}_k}^{n+1}(\hat{c}_k)\}, \quad (2.2)$$

$$\omega_k^{n+1} = \operatorname{argmin}_{\omega_k \in \mathbb{R}^+} \{F_{\omega_k}^{n+1}(\omega_k)\}, \quad (2.3)$$

Here we define a residual function for the k th mode at iteration step n

$$\hat{r}_k^n(\mathbf{x}, \omega) \equiv \hat{q}(\mathbf{x}, \omega) - \sum_{k=1}^{K-1} \phi_k^{n+1}(\mathbf{x}) \hat{c}_k^{n+1}(\omega) - \sum_{k=K+1}^K \phi_k^n(\mathbf{x}) \hat{c}_k^n(\omega) \quad (2.4)$$

Sub-optimization w.r.t. $\phi_k(x)$

$$\phi_k^{n+1}(\mathbf{x}) = \arg \min \left\{ \int_{\Omega} \int_{-\infty}^{\infty} \left| \hat{r}_k^n(\mathbf{x}, \omega) - \phi_k(\mathbf{x}) \hat{c}_k^n(\omega) \right|^2 W(\mathbf{x}) d\omega d\mathbf{x} \right\}. \quad (2.5)$$

Let the functional derivative vanish for all variations

$$\frac{d}{d\delta} \left[\int_{\Omega} \int_{-\infty}^{\infty} \left| \hat{r}_k^n(\mathbf{x}, \omega) - \phi_k(\mathbf{x}) \hat{c}_k^n(\omega) \right|^2 W(\mathbf{x}) d\omega d\mathbf{x} \right] \Big|_{\delta=0} = 0, \quad \forall \psi(\mathbf{x}) \in \Phi, \delta \in \mathbb{R} \quad (2.6)$$

then we have

$$0 = \int_{\Omega} \int_{-\infty}^{\infty} \psi(\mathbf{x}) \hat{c}_k^n(\omega) [\overline{\hat{r}_k^n(\mathbf{x}, \omega) - \phi_k(\mathbf{x}) \hat{c}_k^n(\omega)}] W(\mathbf{x}) d\omega d\mathbf{x} \quad (2.7)$$

$$= \int_{-\infty}^{\infty} \hat{c}_k^n(\omega) [\overline{\hat{r}_k^n(\mathbf{x}, \omega) - \phi_k(\mathbf{x}) \hat{c}_k^n(\omega)}] d\omega \quad (2.8)$$

$$\Rightarrow \int_{-\infty}^{\infty} \hat{c}_k^n(\omega) \overline{\hat{r}_k^n(\mathbf{x}, \omega)} d\omega = \overline{\phi_k(\mathbf{x})} \int_{-\infty}^{\infty} |\hat{c}_k^n(\omega)|^2 d\omega \quad (2.9)$$

$$\Rightarrow \phi_k(\mathbf{x}) = \frac{\int_{-\infty}^{\infty} \overline{\hat{c}_k^n(\omega)} \hat{r}_k^n(\mathbf{x}, \omega) d\omega}{\int_{-\infty}^{\infty} |\hat{c}_k^n(\omega)|^2 d\omega} \quad (2.10)$$

with normalization condition enforced, we have

$$\phi_k^{n+1}(\mathbf{x}) = \frac{\int_{-\infty}^{\infty} \overline{\hat{c}_k^n(\omega)} \hat{r}_k^n(\mathbf{x}, \omega) d\omega}{\left\| \int_{-\infty}^{\infty} \hat{c}_k^n(\omega) \hat{r}_k^n(\mathbf{x}, \omega) d\omega \right\|_{\mathbf{x}, W}} \quad (2.11)$$

Sub-optimization w.r.t. $\hat{c}_k(\omega)$

$$\hat{c}_k^{n+1}(\omega) = \arg \min \left\{ \alpha \int_{-\infty}^{\infty} (|\omega| - \omega_k^n)^2 |\hat{c}_k(\omega)|^2 d\omega + \int_{\Omega} \int_{-\infty}^{\infty} \left| \hat{r}_k^n(\mathbf{x}, \omega) - \phi_k^{n+1}(\mathbf{x}) \hat{c}_k(\omega) \right|^2 W(\mathbf{x}) d\omega d\mathbf{x} \right\}. \quad (2.12)$$

$$\begin{aligned} 0 &= \alpha \int_{-\infty}^{\infty} (|\omega| - \omega_k^n)^2 \overline{\hat{c}_k(\omega)} \eta(\omega) d\omega \\ &\quad + \int_{\Omega} \int_{-\infty}^{\infty} [-\phi_k^{n+1}(\mathbf{x}) \eta(\omega)] \overline{[\hat{r}_k^n(\mathbf{x}, \omega) - \phi_k^{n+1}(\mathbf{x}) \hat{c}_k(\omega)]} W(\mathbf{x}) d\omega d\mathbf{x} \end{aligned} \quad (2.13)$$

$$0 = \alpha (|\omega| - \omega_k^n)^2 \overline{\hat{c}_k(\omega)} - \int_{\Omega} \phi_k^{n+1}(\mathbf{x}) \overline{[\hat{r}_k^n(\mathbf{x}, \omega) - \phi_k^{n+1}(\mathbf{x}) \hat{c}_k(\omega)]} W(\mathbf{x}) d\mathbf{x} \quad (2.14)$$

$$\Rightarrow \hat{c}_k^{n+1}(\omega) = \frac{\int_{\Omega} \overline{\phi_k^{n+1}(\mathbf{x})} \hat{r}_k^n(\mathbf{x}, \omega) W(\mathbf{x}) d\mathbf{x}}{1 + \alpha (|\omega| - \omega_k^n)^2} \quad (2.15)$$

Sub-optimization w.r.t. ω_k

$$\omega_k^{n+1} = \frac{\int_{-\infty}^{\infty} |\omega| |\hat{c}_k^{n+1}(\omega)|^2 d\omega}{\int_{-\infty}^{\infty} |\hat{c}_k^n(\omega)|^2 d\omega} \quad (2.16)$$